NORMAN BUI

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EDUCATION

Carleton University | *Bachelor of Computer Science (Honours),* ***3.9/4.0 GPA*** Graduated Spring 2026

- Coursework: Data Structures & Algorithms, Operating Systems, OOP, Programming Paradigms, DBMS
- Awards: Dean's Honour List every year; President's, Henry Marshall Tory, and Harry S. Southam Scholarships

EXPERIENCE

Canadian Tire Corporation | *Software Developer* July 2026 - Present

- Built an AI tool that reads a codebase in any of **12** languages and searches Confluence for extra context, then writes documentation tracing every fact to the line of code or page it came from
- Rewrote it from TypeScript to Rust (**100k+** lines, **1,100+** tests) with output identical to the original; shipped as a GitHub Copilot plugin that opens Azure DevOps pull requests
- Designed a multi-agent step that ranks a codebase's most important parts before writing, plus a non-AI check that fixes or drops any citation pointing past its code (**10** broken to **0** of **2,100**)
- Made re-runs update only the pages whose code changed, keeping reviewers' edits and cutting re-run cost **40%**
- Cut AI cost per run **38x** by testing candidate models on real projects, with no loss in accuracy

Electronic Arts (EA) | *Software Developer Intern* May 2025 - Aug 2025

- Raised CI/CD load-test pass rate from **35% to 100%** by root-causing backend auth failures and test machine setup
- Integrated **10+** game features into the Kotlin load-testing framework with per-feature metrics
- Helped build an LLM-driven playtesting system that plays the game to find bugs; measured model behavior and built foundations for better soak testing

Autodesk | *Software Developer Intern* Jan 2025 - Apr 2025

- Built TypeScript/Java validation and repair infrastructure for Fusion manufacturing data, including an **O(n)** repair algorithm with concurrency controls that raised throughput **8x**
- Shipped GraphQL validation and repair endpoints with **100%** test coverage (Mocha, Chai, Sinon)
- Made repairs run automatically, catching lost design links before anyone reported them, and closed permission gaps in workflows handling sensitive customer data

Kongsberg Geospatial | *Software Developer Intern* May 2024 - Aug 2024

- Built client-facing C++/Qt geospatial apps for aviation, defense, and government as the sole developer
- Integrated LiDAR, WMTS, and S-57 data into a real-time mapping and object-tracking pipeline
- Optimized the render pipeline to a consistent **60+ FPS** and cut terrain processing time **40%+**

NAV CANADA | *Software Developer Intern* May 2023 - Aug 2023

- Optimized a C++/Qt GIS tool, cutting storage **60%** and speeding up analysis **30%**
- Fixed rendering bugs in air traffic control software, raising frame rate **16%**
- Automated network provisioning with PowerShell, eliminating **95%+** of manual configuration

PROJECTS

Latch | *Electron, TypeScript, Swift, Chrome Extension, Native Messaging* github.com/nbui23/latch

- Built a local-first macOS focus blocker: Electron UI, Swift privileged helper, Chromium extension, native messaging
- Built crash recovery so a focus session keeps blocking through an app crash or browser restart
- Tested it against all **7** recovery cases, plus repair of a broken install

Orbital Refuelling Simulator | *Python, Streamlit, scikit-learn, pytest* github.com/nbui23/orbital-refuelling

- Modeled a **410-second** refuelling mission: **9** phases, **9** injectable anomaly classes, deterministic replay
- Separated hard-limit engineering rules from **9** phase-aware IsolationForest drift models
- Ranked the telemetry channels behind each alert with perturbation-based attribution

SKILLS

Languages: TypeScript, Rust, Python, Java, Kotlin, C++, C#, JavaScript, SQL, Swift, PHP
Systems & Frameworks: Node.js, Next.js, Express.js, Electron, Qt, React Native, Spring Boot, FastAPI, Flask, .NET
LLMs & Agents: GitHub Copilot SDK, multi-agent orchestration, LLM-as-judge evals, RAG, LangGraph, tree-sitter
Data & APIs: PostgreSQL/PostGIS, MySQL, SQLite, MongoDB, Neo4j, Prisma, GraphQL, REST
Infrastructure: Azure DevOps Pipelines, GitHub Actions, Docker, AWS, Linux, Git, CI/CD, TDD (Vitest, pytest, Mocha)
Spoken Languages: English, French, Vietnamese